Which Model Changes Can Counterfactual Explanations See? A Ground-Truth Benchmark

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Abstract. Counterfactual methods are used to explain how a classifier changed after an update, but without ground truth one cannot tell which method explains which kind of change. We inject known changes into classifiers on rendered images with exact factors (a spurious shortcut, a shifted threshold, a subgroup label flip) next to null updates, and compare counterfactual differencing, gradient differences, importance differences, disagreement search and differential counterfactuals (Δ-CE), minimal inputs on which exactly one model changes its decision by a margin. No family wins everywhere, and a pre-registered hypothesis in favour of Δ-CE was rejected. Two findings were confirmed on a fresh test set: gradient differences are nearly blind to a shifted threshold (Recovery 0.10 vs. 0.41, difference 95% CI [0.27, 0.35]), and margin witnesses abstain on null updates, where counterfactual differencing attributes a change to about 99% of points. A linear analysis separating rotation from offset of the decision boundary explains the threshold result but not the shortcut result.

1 Introduction

Deployed classifiers are updated routinely, and an update that improves accuracy can still change individual decisions in unwanted ways [11]. Explaining *how* the decisions of f_{new} differ from those of f_{old} is therefore a natural goal of explainable AI under distributional change [3, 12]. Artelt et al. [1, 2] compare a counterfactual [8] of the same input under each model, or approximate this by a difference of gradients; Muschalik et al. [4] compare feature importances; disagreement search [5, 6, 7] looks for inputs on which two models disagree. These families assume different things about *what* changed (a direction of the weights, a boundary, a region of disagreement), and without ground truth these assumptions go unchecked.

We ask which kind of change each family can see. We inject known changes into classifiers trained on rendered images with exact generating factors and measure which factor each method attributes the change to. We also include *differential counterfactuals* (Δ-CE): the closest input on the data manifold at which only f_{new} changes its decision while f_{old} keeps it, both by a margin κ. Δ-CE anchors disagreement search [5, 6] at a query; unlike robust counterfactuals [9, 10], which seek a point valid for both models, it seeks one valid for exactly one, and the margin makes it return nothing when two models differ only by jitter.

Contributions. (i) A benchmark with exact ground truth for explaining model updates: five rendered factors, three injected changes and two null updates (17 model pairs), with code and protocol. (ii) A comparison of five explanation

families under a protocol fixed before the test data were used, reporting a rejected pre-registered hypothesis, a failed prediction and two confirmed findings. (iii) A linear analysis that separates a rotation of the decision boundary from a shift of its offset, with basic results for the margin witness.

2 Differential counterfactuals and what each family sees

Let $D : \mathcal{Z} \to \mathcal{X}$ map a search space onto the data manifold (here an exact renderer from factor space $\mathcal{Z} = [0,1]^5$). For class y the margin is $m(z) = f_y(D(z)) - f_{1-y}(D(z))$; m_o, m_n belong to f_{old}, f_{new}, and $h = m_n - m_o$. A query z_0 is valid if $m_o(z_0) \geq \kappa$ and $m_n(z_0) \geq \kappa$.

Definition 1. With $\mathcal{F}_\kappa^N = \{z : m_n(z) \leq -\kappa,\ m_o(z) \geq \kappa\}$, a Δ-CE of z_0 is $z^\star \in \arg\min_{z \in \mathcal{F}_\kappa^N} \|z - z_0\|$ if $\|z^\star - z_0\| \leq r$, and NOT_FOUND otherwise; swapping the models gives type O. Each $z \in \mathcal{F}_\kappa^N$ is a *witness*: two forward passes verify that the models decide differently, both decisively, which a difference of two counterfactuals does not.

Lemma 1. Every witness has $h(z) \leq -\kappa - m_o(z) \leq -2\kappa$. Hence models whose margins differ by less than 2κ everywhere have no witness, and if h is L_h-Lipschitz no witness lies within $(h(z_0) + 2\kappa)/L_h$ of z_0. For deep networks L_h is unknown, so this motivates the design rather than certifying it.

Linear case. For linear margins with independent w_o, w_n the witness is unique: the ordinary counterfactual of f_{new} or the minimum-norm point with both constraints active. Write $\hat{w}_n = \cos\theta\, \hat{w}_o + \sin\theta\, v$ with $v \perp \hat{w}_o$, and let d_n, d_o be the distances from z_0 to $\{m_n = -\kappa\}$ and $\{m_o = \kappa\}$. The gradient difference of [1] is $-(\hat{w}_n - \hat{w}_o)$, counterfactual differencing is $u_n - u_o$, the witness is $u^\star$.

Proposition 2. (a) *Offset* ($\theta = 0$): the gradient difference is 0; counterfactual differencing is a multiple of $\hat{w}_o$; a witness exists iff $d_n \leq d_o$ and equals $-d_n \hat{w}_o$. (b) *Rotation* ($0 < \theta < \pi/2$): the gradient difference has components in ratio $\tan(\theta/2) < 1$ along $(\hat{w}_o, v)$, i.e. mainly along the new dependence v; the witness has $|u^\star \cdot \hat{w}_o| = \min\{d_n \cos\theta, d_o\}$ and, for $\theta \leq \pi/4$, is dominated by v only if $d_o(\sin\theta + \cos\theta) < d_n$. After projecting out $\hat{w}_o$ all three coincide, as in Theorem 1 of [1].

Proofs will be released with the code; a numerical check on linear models agrees with the solver on distances to within 10^{-3}, although a stricter pre-declared vector criterion failed at $p = 5$. In words, gradient differences see a *rotation* of the boundary and are blind to a shift of its *offset*; counterfactual differencing and witnesses see the offset, and witnesses abstain when the shift is below 2κ. This analysis was formulated after the main experiment.

3 Benchmark and protocol

Data and models. Images are 32×32 ellipses rendered from five independent factors in $[0,1]$: aspect (task factor), size, colour, background and horizontal position; labels are $y = \mathbb{1}[\text{aspect} > 0.5]$ with 5% noise. The renderer is exact, so the factor change of any explanation is read without a learned oracle. f_{old} is

Table 1: Injected changes; $k^\star$ is the ground-truth factor.

Code	Adaptation data	$k^\star$	Pairs
T1 shortcut	colour correlated with label (0.99), near the boundary	colour	3
T3 subgroup	labels flipped for size > 0.75	size	3
T4 threshold	$y = \mathbb{1}$[aspect > 0.6]	aspect	3
N0 / N1 null	resample / size $\sim U[0.5, 1]$, same rule	—	5 / 3

Table 2: Recovery@r (found-rate) on the locked test split, mean over three pairs; all 95% bootstrap CIs within ± 0.035. Last row: found-rate on the 8 null pairs.

	Δ-CE	B1	B1-det	B3	B4	B5	A1	A2
T1	0.38 (0.84)	0.54 (0.93)	0.54 (0.93)	**0.91** (1.0)	0.00	0.02 (0.08)	0.35 (0.63)	0.07 (0.93)
T3	0.53 (0.81)	0.32 (1.0)	0.32 (0.99)	**0.62** (1.0)	0.00	0.06 (0.10)	0.29 (0.58)	0.30 (1.0)
T4	0.47 (0.47)	0.77 (0.90)	0.76 (0.90)	0.07 (1.0)	**1.00**	0.07 (0.08)	0.31 (0.40)	0.90 (0.90)
Null	**0.000**	0.995	0.994	1.0	1.0	0.000	0.000	0.996

a small CNN (five seeds); f_{new} is f_{old} fine-tuned on 10 000 samples from one of five conditions (Table 1). A pair enters only if its change is visible in decisions on validation data (pre-declared thresholds on decision flips per factor; null pairs need decisive disagreement ≤ 0.02); all 17 qualified.

Metrics and methods. Per pair we use $N = 300$ valid test points. Recovery@r is the share of points with a valid explanation within the budget whose largest factor change is $k^\star$ (NOT_FOUND and ties count 0). All counterfactual methods share one solver (Adam with a decaying penalty, then SLSQP on the constrained problem), the same points and the same budget $r = 0.54$, set once on validation. We compare Δ-CE ($\kappa = 0.5$); **B1** counterfactual differencing [1] and a deterministic variant **B1-det**; **B3** gradient difference [1]; **B4** permutation-importance difference [4]; **B5** DIG-style line search [5]; **A1** DeepXplore-style search without proximity [6]; **A2** the ordinary counterfactual of f_{new} alone.

Protocol. All choices were fixed before the test split was used once. *Pre-registered:* H2a (Δ-CE beats B3 on T3; paired bootstrap stratified by pair, 95% CI, all three seeds) and a tie on T1 and T4 ($|\Delta| \leq 0.10$). *Confirmatory:* readings formed after the results, tested on a new test set with the same models (C1–C3; Holm correction over C1 and C3). *Descriptive:* the rest. A second tier on coloured digits and a label-shift update failed pre-declared qualification and were stopped.

4 Results

Pre-registered. H2a is *rejected*: on T3, Recovery(Δ-CE) − Recovery(B3) $= -0.091$, 95% CI $[-0.133, -0.050]$, negative in all seeds. Δ-CE beats B1 (+0.21, descriptive) and B5 (+0.47, CI [0.43, 0.50]), which rarely reaches the disagreement region (found-rate ≤ 0.11); we make no solver claim from one benchmark. The predicted tie also fails: all four comparisons exceed 0.10 (T1 vs. B3 −0.53, vs.

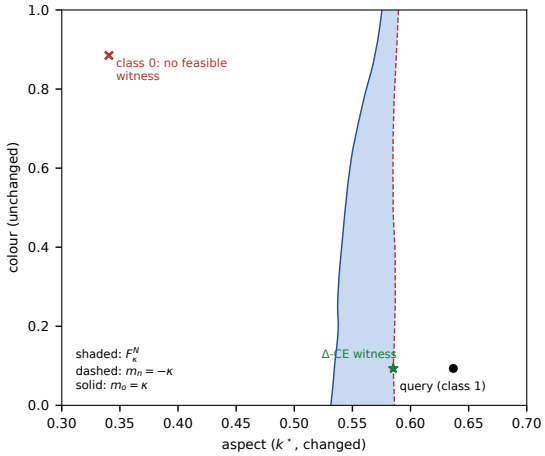


Fig. 1: T4 (one pair) in the (aspect, colour) plane, the other three factors fixed at the witness's values (they differ from the query's by at most 0.005 and from the class-0 point's by at most 0.024). Shaded: witness region $\mathcal{F}^N_\kappa$, bounded by $m_n = -\kappa$ (dashed) and $m_o = \kappa$ (solid). A class-1 query, its Δ-CE witness on the edge of the region, and a class-0 point with no feasible witness. Illustrative, not a result.

B1-det −0.17; T4 vs. B3 +0.40, vs. B1-det −0.29). B1 and B1-det agree within 0.01 everywhere.

Null updates. On all eight null pairs Δ-CE returns no witness, while B1 attributes a change to 99.5% of points. With $\kappa = 0$ witnesses reappear (found-rate 0.80 on N0, 0.89 on N1), as Lemma 1 predicts. All methods separate drift from null pairs at the pair level (AUROC 1.0), so the difference lies in what each says about a null update.

Confirmed on a new test set. *C1 (offset blindness), confirmed:* on T4, Recovery(Δ-CE) − Recovery(B3) = +0.31, 95% CI $[0.27, 0.35]$, $p_{Holm} = 2 \cdot 10^{-4}$, positive in all seeds (Δ-CE 0.41, B3 0.10, B1-det 0.76), the CNN counterpart of Proposition 2(a). Its content is the failure of B3: on an offset change the changed factor is the task factor, so even A2 recovers it (0.90). *C2 (abstention), confirmed:* no witness on any null pair; B1 attributes a change to 99.1% (N0) and 99.8% (N1) of points. *C3 (rotation after projection), not confirmed:* Proposition 2(b) predicts that Δ-CE and B3 coincide on T1 once $\hat{w}_o$ is projected out; they do not (0.64 vs. 0.95, difference −0.30, 90% CI $[-0.33, -0.28]$). The T1 outcome remains unexplained.

Where witnesses exist (descriptive). On T4 a witness is found for every class-1 point and for no class-0 point, in all pairs and on both test sets (e.g. 136/136 vs. 0/164); class-0 failures are infeasible, not over budget (Fig. 1). This is what Proposition 2(a) implies for an offset that lowers the margin of one class, and the class-wise statement was written before the split was computed. It is

what a witness adds on T4 beyond the direction: the side of the boundary on which decisions flipped. On T1 and T3, 16–17% of points have a witness beyond the budget; halving the budget halves the found-rate.

Real digits (exploratory). The tier on coloured MorphoMNIST digits (4 vs. 9, decoded from a 16-dimensional VAE latent) failed qualification for the injected changes, but its five null pairs could still be run. Δ-CE returned no witness on four of them; on the fifth, whose test disagreement rate was the highest (1.2%, about three times the mean of the others), it found witnesses for 47.5% of points (25% of type N). B1 attributed a change to 97–99% of points on every pair and concentrated it on one factor on all five (systematic attribution rate 1.0, against 0 for type-N witnesses). This is one exploratory analysis on five pairs, so we read it only as consistent with the abstention result.

5 Discussion and limitations

Which family explains a model change depends on the kind of change. Gradient differences recover a change of direction (T1) and are nearly blind to a change of offset (T4). Counterfactual differencing cannot say that nothing changed. Margin witnesses are the only family that abstains on null updates and returns checkable inputs, but they are limited by feasibility and budget and lose on the subgroup change. A reading not tested as a procedure is to use witnesses to decide *whether* decisions changed near a query and other families to describe *how*.

Limitations. (1) The data are synthetic; the digit tier failed qualification because fine-tuning a strong model no longer produced the injected changes, so only its null pairs are reported. (2) Proposition 2(a) implies that the update optimal under label shift, a pure offset [13], is invisible to gradient differences, but our label-shift update created no decisive disagreements, so this remains open. (3) T1 is unexplained (C3). (4) Witnesses see only decision changes on the manifold within the budget, with no certificate for deep networks. (5) C1–C3 were formed after the main experiment. (6) On offset changes Recovery rewards any explanation that follows f_{new} (A2: 0.90), so it measures blindness rather than sensitivity; a metric that scores *where* an explanation is returned is needed. The protocol went through eight logged amendments and two pre-declared stopping rules; history, proofs and code will be released with the camera-ready version.

References

[1] A. Artelt, F. Hinder, V. Vaquet, R. Feldhans, B. Hammer. Contrastive explanations for explaining model adaptations. arXiv:2104.02459, 2021.

[2] A. Artelt et al. Contrasting explanations for understanding and regularizing model adaptations. *Neural Processing Letters*, 2022. doi:10.1007/s11063-022-10826-5.

[3] B. Hammer, E. Hüllermeier. Interpretable machine learning: on the problem of explaining model change. In *Proc. 31. Workshop Computational Intelligence*, pp. 1–10, KIT Scientific Publishing, 2021.

[4] M. Muschalik, F. Fumagalli, B. Hammer, E. Hüllermeier. Agnostic explanation of model change based on feature importance. *KI – Künstliche Intelligenz* 36:211–224, 2022.

[5] X. Renard, T. Laugel, M. Detyniecki. Understanding prediction discrepancies in machine learning classifiers. arXiv:2104.05467, 2021.

[6] K. Pei, Y. Cao, J. Yang, S. Jana. DeepXplore: automated whitebox testing of deep learning systems. *SOSP*, 2017.

[7] M. Augustin et al. DiG-IN: diffusion guidance for investigating networks. *CVPR*, 2024.

[8] S. Wachter, B. Mittelstadt, C. Russell. Counterfactual explanations without opening the black box. *Harvard J. Law & Technology*, 2017.

[9] F. Hamman et al. Robust counterfactual explanations for neural networks with probabilistic guarantees. *ICML*, 2023.

[10] M. Pawelczyk, K. Broelemann, G. Kasneci. On counterfactual explanations under predictive multiplicity. *UAI*, 2020.

[11] S. Yan, Y. Xiong, K. Kundu, S. Yang, S. Deng, M. Wang, W. Xia, S. Soatto. Positive-congruent training: towards regression-free model updates. *CVPR*, 2021.

[12] M. Kaden, B. Paassen, B. Hammer, R. Schubert, T. Villmann. Reliable counterfactuals for machine learning models — current aspects and perspectives. *ESANN*, 2026.

[13] M. Saerens, P. Latinne, C. Decaestecker. Adjusting the outputs of a classifier to new a priori probabilities: a simple procedure. *Neural Computation* 14(1):21–41, 2002.